Product Data Cleaning & Transformation

Work Description / Project Summary

Project Title: Product Dataset Data Cleaning and Transformation

Work Description: Clean, standardize, transform, and format a product dataset to improve data quality, consistency, usability, and readiness for analysis or reporting.

1. Handling Missing Values

Check missing values in the **Price** column and determine an appropriate treatment, such as removing records where the price cannot be reliably determined or imputing it using a suitable business rule. Identify missing **Category** values and apply an appropriate imputation or 'Unknown' category strategy.

2. Correcting Inconsistent Data

Identify inconsistent text formats in **Product Name** and standardize them using Find and Replace. Detect and correct spelling errors or typos in the **Category** column to maintain consistent categories.

3. Removing Duplicates

Check the complete dataset for duplicate rows based on all columns and remove duplicate records where identified.

4. Splitting and Merging Data

Split the **Product ID** into separate **Manufacturing Date** and **Country Code** fields, removing unnecessary characters. Merge **Brand Name** and **Product Name** into a new **Product Brand** column.

5. Number & Date Formatting

Convert the **Price** field to currency format and format **Manufacturing Date** as **DD-MM-YYYY** for consistent presentation.

6. Conditional Formatting

Apply data bars or color scales to the **Price** column to visualize price variation. Create a custom rule in the **Category** column to highlight records where the category is **Electronics**.

Key Outcome	A cleaned, standardized, duplicate-free and properly formatted product dataset suitable for further analysis, visualization, and reporting.